

DATA ANALYTICS INTERNSHIP

AN INTERNSHIP PROJECT REPORT

Submitted by

A S SUJEETH

(Reg No: 110725102252)

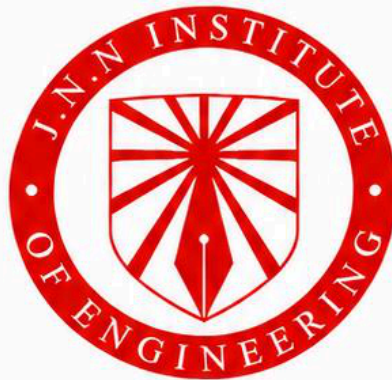
in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

Artificial Intelligence & Data Science



J.N.N Institute of Engineering

(An Autonomous Institute, Affiliated to Anna University, Chennai)

WIZTECH AUTOMATION SOLUTIONS

(05.08.2026 – 20.08.2026)

2026 - 2027



VISION OF THE INTERNSHIP ORGANIZATION:

To be a leading provider of innovative automation and data-driven solutions that empower industries through technology, excellence and continuous improvement, creating smarter and more efficient business operations.

MISSION OF THE INTERNSHIP ORGANIZATION:

IM1: To deliver reliable automation and data analytics solutions tailored to client requirements with a focus on quality and efficiency.

IM2: To leverage advanced technologies and analytical tools to convert data into meaningful insights for better decision making.

IM3: To provide industry-oriented training and hands-on experience to students and young professionals in emerging technologies.

IM4: To foster a culture of innovation, continuous learning and technical excellence in all our projects and services.

IM5: To build long-term relationships with clients, educational institutions and organizations by offering knowledge sharing, practical exposure and real-world solutions.



**J.N.N INSTITUTE OF
ENGINEERING
AUTONOMOUS**

NAAC 'A' Grade | Approved by AICTE | Affiliated to Anna University

Approved by



Accredited by



VISION OF THE INTERNSHIP ORGANIZATION:

To be a leading provider of innovative automation and data-driven solutions that empower industries through technology, excellence and continuous improvement, creating smarter and more efficient business operations.

MISSION OF THE INTERNSHIP ORGANIZATION:

- **DM 1:** To deliver reliable automation and data analytics solutions tailored to client requirements with a focus on quality and efficiency.
- **DM 2:** To leverage advanced technologies and analytical tools to convert data into meaningful insights for better decision making.
- **DM 3:** To provide industry-oriented training and hands-on experience to students and young professionals in emerging technologies.
- **DM 4:** To foster a culture of innovation, continuous learning and technical excellence in all our projects and services.
- **DM 5:** To build long-term relationships with clients, educational institutions and organizations by offering knowledge sharing, practical exposure and real-world solutions.



**J.N.N INSTITUTE OF
ENGINEERING**
AUTONOMOUS

NAAC 'A' Grade | Approved by AICTE | Affiliated to Anna University

Approved by



Accredited by



PROGRAM OUTCOMES (POs)

S.No	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply knowledge of mathematics, statistics, computing, artificial intelligence, data science and engineering fundamentals to analyze and solve real-world problems in the field of data-driven technologies.
PO2	Problem analysis	Identify, formulate, review research literature and analyze complex problems in Artificial Intelligence and Data Science using analytical and critical thinking skills, reaching well-substantiated conclusions.
PO3	Design/Development of solutions	Design and develop data-driven solutions, models and systems using appropriate tools and techniques in AI and Data Science to meet the needs of society and industry with consideration for ethical, societal, environmental and safety aspects.
PO4	Conduct investigations of complex problems	Use research-based knowledge and analytical methods including data collection, preprocessing, modeling, analysis and interpretation of data to provide valid conclusions for complex engineering and real-world problems.
PO5	Modern tool usage	Create, select and apply modern computing and data science tools and technologies including programming, databases, data analytics, machine learning and visualization tools to solve complex problems with an understanding of their limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the resulting responsibilities relevant to professional practice in Artificial Intelligence and Data Science.
PO7	Environment and sustainability	Understand the impact of AI and data-driven solutions in societal and environmental contexts and demonstrate the knowledge and need for sustainable development practices.



J.N.N INSTITUTE OF ENGINEERING AUTONOMOUS

NAAC 'A' Grade | Approved by AICTE | Affiliated to Anna University

Approved by



Accredited by



PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities in the field of Artificial Intelligence and Data Science, considering the impact of engineering solutions on society, environment and human values.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings to achieve common goals in the field of AI and Data Science.
PO10	Communication	Communicate effectively on complex engineering ideas and AI solutions with the engineering community and society at large, through well-structured reports, documentation, presentations and visualizations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage AI and Data Science projects and in multidisciplinary environments with consideration for cost, resources and time.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change in Artificial Intelligence and Data Science.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To equip graduates with strong foundation in mathematics, computing, Artificial Intelligence and Data Science to analyze, design and develop data-driven solutions for real-world problems.
PEO2	To prepare graduates to pursue higher education, research and professional careers in the field of AI and Data Science with the ability to adapt to emerging technologies and industry needs.
PEO3	To develop socially responsible and ethically sound professionals who contribute to the society through innovative, sustainable and impactful AI and Data Science solutions.



**J.N.N INSTITUTE OF
ENGINEERING
AUTONOMOUS**

NAAC 'A' Grade | Approved by AICTE | Affiliated to Anna University

Approved by



Accredited by



PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	To develop domain-specific expertise in Artificial Intelligence and Data Science by applying advanced analytical techniques, machine learning models and data-driven approaches to solve real-world problems in diverse domains such as healthcare, business, industry and society.
PSO2	To enable graduates to design, develop and deploy intelligent systems and data-driven solutions using modern tools and technologies with an emphasis on innovation, problem solving and sustainable practices.
PSO3	To prepare graduates for higher studies, research and professional careers in emerging areas of Artificial Intelligence and Data Science by fostering critical thinking, ethical practices and lifelong learning.



**J.N.N INSTITUTE OF
ENGINEERING
AUTONOMOUS**

NAAC 'A' Grade | Approved by AICTE | Affiliated to Anna University

Approved by



Accredited by



BONAFIDE CERTIFICATE

Certified that this internship project report “**DATA ANALYTICS – SALES DATA ANALYSIS AND VISUALIZATION**” is the bonafide work of “**A S SUJEETH (Reg. No: 110725102252)**” who carried out the project work under my supervision during the academic year 2026 – 2027.

Signature of the HOD with date

Dr. P. Nagarajan

Professor and Head,
Department of Artificial Intelligence
& Data Science Engineering,
J.N.N Institute of Engineering,
Chennai – 602 024.

Signature of the Supervisor with date

Miss. M. Gayathridevi

Project Supervisor,
Department of Artificial Intelligence
& Data Science Engineering,
J.N.N Institute of Engineering,
Chennai – 602 024.

Certified that the above students have attended of viva voce during the exam
held on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, I express my sincere gratitude to the management of **J.N.N Institute of Engineering** for providing the academic support and facilities required to complete my internship report.

I sincerely thank the Principal and the Head of the Department of **Artificial Intelligence & Data Science** for their constant encouragement, valuable guidance and support throughout the internship period.

I express my heartfelt thanks to **WIZTECH AUTOMATION SOLUTIONS, Chennai**, for providing me with the opportunity to undergo the Data Analytics internship from **05.08.2026 to 20.08.2026** and to work on the project titled “**Sales Data Analysis and Visualization**”.

I would also like to thank my internship supervisor **Miss. M. Gayathridevi**, for her continuous guidance, valuable suggestions and support provided during the preparation and completion of this report.

Finally, I thank my parents, friends, faculty members, and everyone who supported and encouraged me throughout the internship.

ABSTRACT

The Data Analytics internship at **WIZTECH AUTOMATION SOLUTIONS, Chennai**, provided practical exposure to the process of analyzing and visualizing sales data. The internship project, titled “**Sales Data Analysis and Visualization**”, was carried out from **05.08.2026 to 20.08.2026**.

The project focused on understanding sales data, preparing the data for analysis, identifying meaningful patterns and trends, and presenting the findings through suitable visualizations. The work involved a structured data-analysis workflow beginning with data collection, understanding the dataset and its attributes, followed by data cleaning, preprocessing, exploratory data analysis, calculation of relevant sales measures and key performance indicators, and visualization of the results using appropriate tools.

The internship helped develop a practical understanding of how data can be transformed into useful information for interpretation and decision-making. It also provided experience in organizing an analytical workflow, selecting appropriate visual representations, interpreting sales-related patterns, and documenting the observations obtained from the analysis. This project enhanced my analytical skills and strengthened my knowledge in data analytics and visualization.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
1	INTRODUCTION	13
	1.1 OVERVIEW	13
	1.2 INTERNSHIP DETAILS	13
	1.3 OBJECTIVES	13
	1.4 SCOPE OF THE INTERNSHIP	13
2	LITERATURE REVIEW	15
	2.1 DATA ANALYTICS	15
	2.2 DATA PREPARATION	15
	2.3 EXPLORATORY DATA ANALYSIS	15
	2.4 DATA VISUALIZATION	15
3	METHODOLOGY	17
	3.1 OVERALL APPROACH	17
	3.2 DATA UNDERSTANDING	17
	3.3 DATA PREPARATION	17
	3.4 EXPLORATORY ANALYSIS	17
	3.5 VISUALIZATION AND INTERPRETATION	17
4	IMPLEMENTATION	19
	4.1 PROJECT SETUP AND DATA UNDERSTANDING	19
	4.2 DATA CLEANING AND PREPROCESSING	19
	4.3 SALES DATA EXPLORATION	19
	4.4 VISUALIZATION OF SALES DATA	19
	4.5 INTERPRETATION OF RESULTS	19
	4.6 TOOLS AND TECHNOLOGIES USED	21
	4.7 IMPLEMENTATION STEPS	21
	4.8 SAMPLE OUTPUT	21
	4.9 SUMMARY	21
5	RESULTS AND DISCUSSION	23
	5.1 DATA QUALITY AND PREPARATION	23
	5.2 DATA ANALYSIS	23
	5.3 CATEGORY AND ATTRIBUTE ANALYSIS	23
	5.4 VISUALIZATION-BASED OBSERVATIONS	23
	5.5 LEARNING OUTCOMES	23
	5.6 KEY FINDINGS	25
	5.7 DISCUSSION	25
	5.8 CHALLENGES FACED	25
	5.9 OVERALL DISCUSSION (CONTD.)	25
6	CONCLUSION AND FUTURE WORK	27
	6.1 CONCLUSION	27
	6.2 OVERALL INTERNSHIP OUTCOME	27
	6.3 FUTURE WORK	27
	6.4 LIMITATIONS OF THE STUDY	28
	6.5 SUMMARY	28

LIST OF FIGURES

CHAPTER	FIGURE NO.	TITLE OF THE FIGURE	PAGE NO.
2	Fig 1	Sample Visualizations of Sales Data	15
4	Fig 1	Overall project setup workflow	19
4	Fig 2	Preview of the sales dataset	19
4	Fig 3	Sample visualizations of sales data	19
4	Fig 4	Tools and technologies used	21
4	Fig 5	Sample code for data loading	21
4	Fig 6	Implementation workflow	21
4	Fig 7	Summary statistics of the dataset	21
4	Fig 8	Monthly sales trend (sample output)	21

Note: Figures are numbered chapter-wise as they appear in the report. Figure 1 of Chapter 2 and Figures 1 to 8 of Chapter 4 follow the numbering used in the captions.

LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM	APPEARS IN
AI	Artificial Intelligence	Programme details, Ch. 1
AICTE	All India Council for Technical Education	Institution header
B.Tech	Bachelor of Technology	Cover page, certificates
CSV	Comma-Separated Values	Ch. 4 (dataset file)
DS	Data Science	Programme details, Ch. 6
EDA	Exploratory Data Analysis	Ch. 2, 3, 4, 5
HOD	Head of the Department	Bonafide certificate
KPI	Key Performance Indicator	Abstract, Ch. 1
NAAC	National Assessment and Accreditation Council	Institution header
PEO	Program Educational Objectives	Programme details
PO	Program Outcomes	Programme details
PSO	Program Specific Objectives	Programme details

CHAPTER 1 INTRODUCTION

1.1 Overview

Data Analytics is the process of collecting, preparing, analysing and interpreting data to extract meaningful information and support decision-making. Sales data contains valuable information about transactions, products, quantities, prices and customer behavior, which can help businesses identify trends, patterns and opportunities for growth.

During the Data Analytics internship at **WIZTECH AUTOMATION SOLUTIONS, Chennai**, the project titled “**Sales Data Analysis and Visualization**” was undertaken as part of the internship program. The internship was completed from **05.08.2026 to 20.08.2026**. The project followed a systematic workflow covering data understanding, preparation, analysis and visualization.

1.2 Internship Details

Organization	: WIZTECH AUTOMATION SOLUTIONS, Chennai
Internship Domain	: Data Analytics
Project Title	: Sales Data Analysis and Visualization
Internship Period	: 05.08.2026 to 20.08.2026
College	: J.N.N Institute of Engineering, Chennai – 602 024.
Student	: A S Sujeeth
Register Number	: 110725102252
Internship Supervisor	: Miss. M. Gayathridevi

1.3 Objectives

- To understand the structure and characteristics of sales data.
- To learn the process of collecting, cleaning and preparing sales data for analysis.
- To identify useful trends, patterns and relationships in sales information.
- To represent sales data using appropriate charts and visualizations.
- To interpret the analytical results in a clear and meaningful manner.
- To develop practical skills in a structured Data Analytics workflow using modern tools.

1.4 Scope of the Project

The project covers the preparation, analysis and visualization of sales-related data. The scope includes examining key sales fields, identifying and handling data quality issues, deriving useful measures and key performance indicators, analysing sales patterns and trends, and presenting the observations through suitable visualizations. The dataset fields, analytical measures and tool used during the internship are aligned with the project requirements.

CHAPTER 1 AT A GLANCE

Internship Snapshot

Organization	: WIZTECH AUTOMATION SOLUTIONS, Chennai
Domain	: Data Analytics
Project	: Sales Data Analysis and Visualization
Period	: 05.08.2026 to 20.08.2026

Objectives at a Glance

NO.	OBJECTIVE	ADDRESSED IN SECTION	RESULTING OUTPUT
1	Understand the structure and characteristics of sales data	3.2, 4.1	Dataset structure and fields
2	Collect, clean and prepare sales data for analysis	3.3, 4.2	Clean, consistent dataset
3	Identify useful trends, patterns and relationships	3.4, 4.3	Summaries and comparisons
4	Represent sales data using appropriate charts	3.5, 4.4	Line, bar, pie and horizontal bar charts
5	Interpret the analytical results clearly	4.5, 5.6, 5.7	Key findings and discussion
6	Develop practical skills in a structured workflow using modern tools	4.6, 4.7, 5.5	Python workflow in Google Colab

Scope of the Project

Key sales fields	Data quality issues	Measures and KPIs	Patterns and trends	Suitable visualizations
------------------	---------------------	-------------------	---------------------	-------------------------

The scope covers the preparation, analysis and visualization of sales-related data, with the dataset fields, analytical measures and tool aligned to the project requirements.

CHAPTER 2 LITERATURE REVIEW

2.1 Data Analytics

Data Analytics is the process of collecting, processing, analyzing and interpreting data to extract meaningful information and support decision-making. It uses statistical methods, analytical techniques and visualization tools to find patterns, trends and relationships in data. In business, sales data analytics helps in understanding customer behavior, product performance, regional trends and revenue growth, which are useful for strategic planning.

2.2 Data Preparation

Data preparation converts raw data into a clean and structured format suitable for analysis. It includes handling missing values, removing duplicate records, correcting inconsistent formats, standardizing data types and creating the required fields. Proper data preparation improves the quality and reliability of the analysis and helps in reducing errors during subsequent steps.

2.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the main characteristics of a dataset before drawing conclusions. It involves descriptive statistics, summary measures, frequency analysis and graphical methods to examine distributions, trends and relationships among variables. EDA helps in identifying key insights, unusual patterns and potential data issues.

2.4 Data Visualization

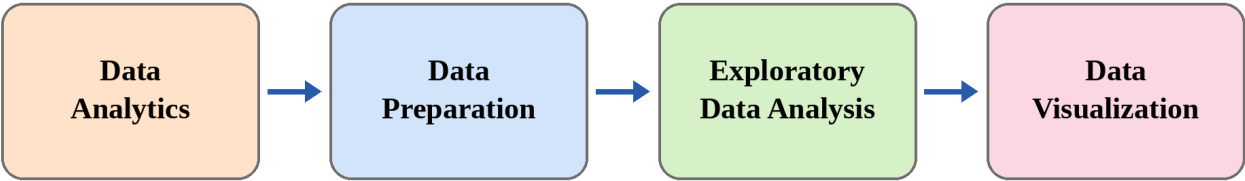
Data visualization represents analytical results in graphical formats that are easy to understand and interpret. Various plots such as line charts, bar charts, pie charts, heatmaps and other suitable visualizations can be used to communicate findings clearly. Visualizations help in comparing values, identifying trends, highlighting important patterns and supporting decision-making.



Fig 1: Sample Visualizations of Sales Data

CHAPTER 2 AT A GLANCE

Concept Flow of the Literature Review



Summary of Concepts

CONCEPT	PURPOSE	METHODS / ELEMENTS	BENEFIT
Data Analytics	Extract meaningful information and support decision-making	Statistical methods, analytical techniques and visualization tools	Understanding customer behavior, product performance, regional trends and revenue growth
Data Preparation	Convert raw data into a clean, structured format suitable for analysis	Handling missing values, removing duplicates, correcting formats, standardizing data types, creating required fields	Improves quality and reliability and reduces errors in later steps
Exploratory Data Analysis	Understand the main characteristics of a dataset before drawing conclusions	Descriptive statistics, summary measures, frequency analysis, graphical methods	Identifies key insights, unusual patterns and potential data issues
Data Visualization	Present results in graphical formats that are easy to interpret	Line charts, bar charts, pie charts, heatmaps	Compares values, shows trends and highlights patterns for decision-making

CHAPTER 3 METHODOLOGY

3.1 Overall Approach

The project followed a systematic and sequential data analytics workflow to derive meaningful insights from the sales dataset. The workflow included data understanding, data preparation, exploratory data analysis, data visualization and interpretation of results. The approach ensured that the data was accurate, well-structured and suitable for analysis, and the findings were presented in a clear and meaningful manner.

3.2 Data Understanding

The first stage involved examining the available sales dataset to understand its structure. The dataset was explored to identify its rows, columns, data types and the meaning of each field. The data dictionary or source information, where available, was referred to understand the business context and significance of the attributes such as date, product, category, quantity, price, region and sales amount.

3.3 Data Cleaning and Preprocessing

The data preparation stage included checking for missing or invalid values, identifying and removing duplicate records, correcting inconsistent formats, standardizing data types and handling outliers. Numerical and categorical fields were refined to ensure that the data was clean, consistent and ready for analysis. Any required transformations, such as date format conversion, category mapping or calculation of new fields, were performed before analysis.

3.4 Exploratory Analysis

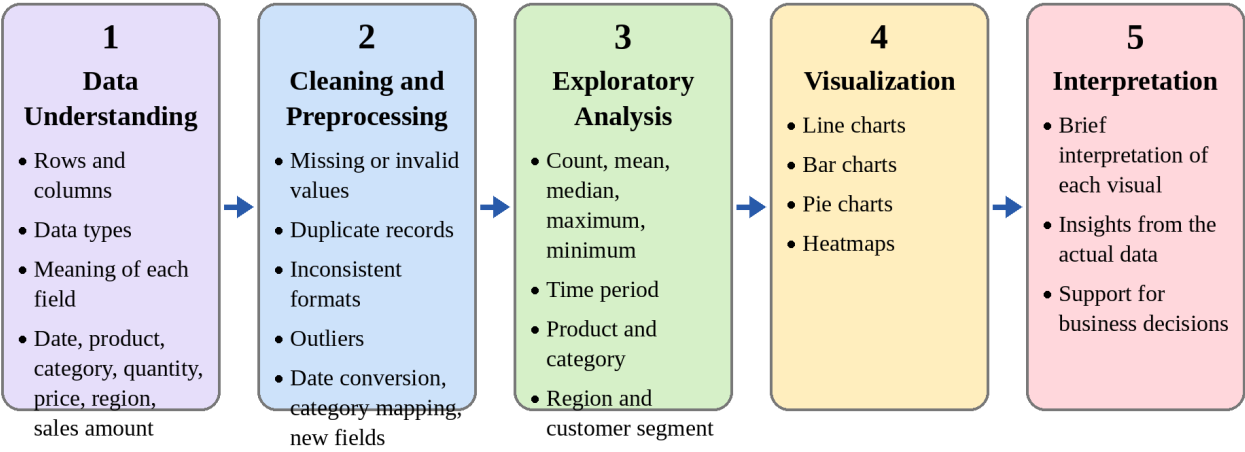
After preprocessing, exploratory data analysis (EDA) was performed to examine the main characteristics of the sales data. Descriptive statistics such as count, mean, median, maximum and minimum were used to understand the distribution of key variables. The analysis was organized around important business dimensions such as time period, product, category, region and customer segment to identify trends, patterns and relationships.

3.5 Visualization and Interpretation

Suitable visualizations were created to represent the analytical findings. Line charts were used to show trends over time, bar charts to compare categories or products, pie charts to show proportions, and heatmaps to identify patterns across regions and months. Each visualization was accompanied by a brief interpretation to explain the insights obtained from the actual data and to support business decision-making.

CHAPTER 3 AT A GLANCE

Stages of the Methodology



Choice of Visualization

CHART TYPE	PURPOSE	USED FOR
Line chart	Show trends over time	Monthly sales trend
Bar chart	Compare categories or products	Category-wise and product-wise sales
Pie chart	Show proportions	Regional distribution of sales
Heatmap	Identify patterns across regions and months	Region-wise monthly sales

CHAPTER 4 IMPLEMENTATION

4.1 Project Setup and Data Understanding

The project began by obtaining the sales dataset used for the internship and examining its structure. The dataset was reviewed to identify the available columns, number of records, data types and the information represented by each attribute. Python libraries such as pandas, matplotlib and seaborn were used in Google Colab for data analysis.

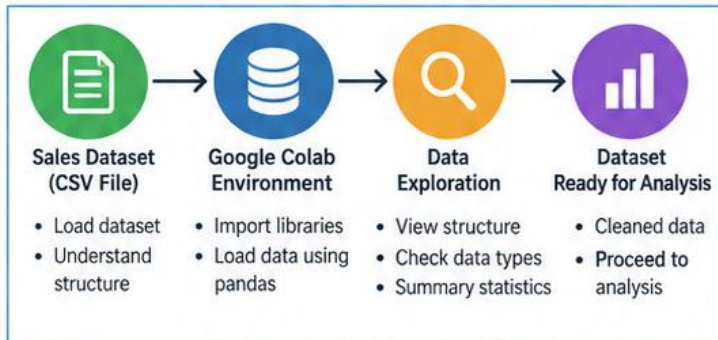


Fig 1: Overall project setup workflow

	Order_ID	Date	Product	Category	Quantity	Price	Region	Sales
0	1001	2023-01-05	Laptop	Electronics	2	55000	South	110000
1	1002	2023-01-08	Mouse	Electronics	5	800	South	4000
2	1003	2023-01-12	Chair	Furniture	3	2500	West	7500
3	1004	2023-01-15	Table	Furniture	1	7000	North	7000
4	1005	2023-01-18	Monitor	Electronics	4	12000	East	48000

Fig 2: Preview of the sales dataset

4.2 Data Cleaning and Preprocessing

The raw data was checked for missing values, duplicate entries, inconsistent values and formatting issues. The required fields were converted into appropriate formats, such as date format conversion, categorical encoding and numeric standardization, so that the analysis could be performed consistently.

4.3 Sales Data Exploration

The prepared dataset was explored using descriptive analysis. Sales-related measures such as total sales, average sales, top products and regional performance were summarized and compared across different dimensions. The analysis focused on identifying trends, patterns and high- and low-performing segments in the sales data.

4.4 Visualization of Sales Data

Visualizations were prepared to communicate the analytical observations clearly. Time-based sales information was represented using line charts, category-wise comparison using bar charts, regional distribution using pie charts and product performance using a horizontal bar chart.



Fig 3: Sample visualizations of sales data

4.5 Interpretation of Results

The visualizations and analytical summaries were reviewed together to identify meaningful observations. Interpretations were based only on the patterns visible in the prepared dataset. The final presentation results included the actual charts and numerical values generated during the internship.

CHAPTER 4 AT A GLANCE : DATASET

Dataset Fields

COLUMN	DESCRIPTION	DATA TYPE	EXAMPLE (FIG 2)
Order_ID	Unique order identifier	Integer	1001
Date	Date of the sale	Date	2023-01-05
Product	Product sold	Text	Laptop
Category	Product category	Text	Electronics
Quantity	Number of units sold	Integer	2
Price	Unit price of the product	Numeric	55000
Region	Region of the sale	Text	South
Sales	Sales amount for the order	Numeric	110000

Dataset Summary (Fig 7)

METRIC	VALUE
Total Sales ()	3,65,000
Average Sales ()	7,300
Total Products	5
Total Records	50
Date Range	01-Jan-2023 to 31-Dec-2023

Observation from the Preview

In every previewed record, the Sales value equals Quantity multiplied by Price (for example, $2 \times 55000 = 1,10,000$ for Order 1001), showing that Sales is a derived field built from the other two columns.

CHAPTER 4 IMPLEMENTATION (Contd.)

4.6 Tools and Technologies Used

The project was implemented using Python in Google Colab. Various libraries were used for data manipulation, analysis and visualization. The tools helped in performing data cleaning, exploratory analysis and creating meaningful visualizations.

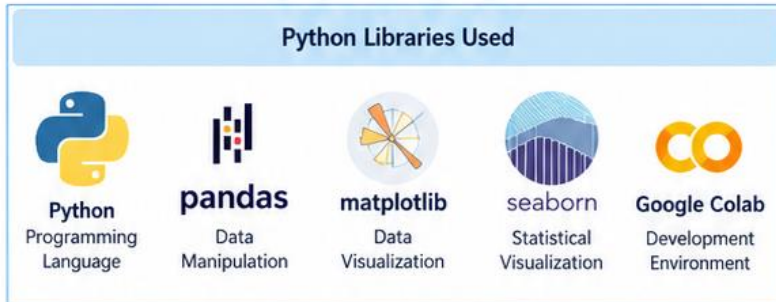


Fig 4: Tools and technologies used

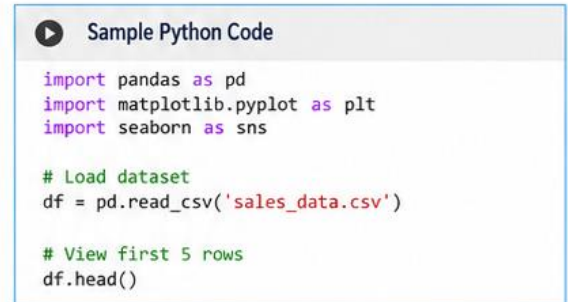


Fig 5: Sample code for data loading

4.7 Implementation Steps

The project followed a systematic workflow from data collection to visualization and interpretation.



Fig 6: Implementation workflow

4.8 Sample Output

Some of the key outputs obtained during the implementation include data summaries, statistical analysis and visualizations showing sales trends, category-wise performance and regional comparison.

Data Summary	
Metric	Value
Total Sales (₹)	3,65,000
Average Sales (₹)	7,300
Total Products	5
Total Records	50
Date Range	01-Jan-2023 to 31-Dec-2023

Fig 7: Summary statistics of the dataset



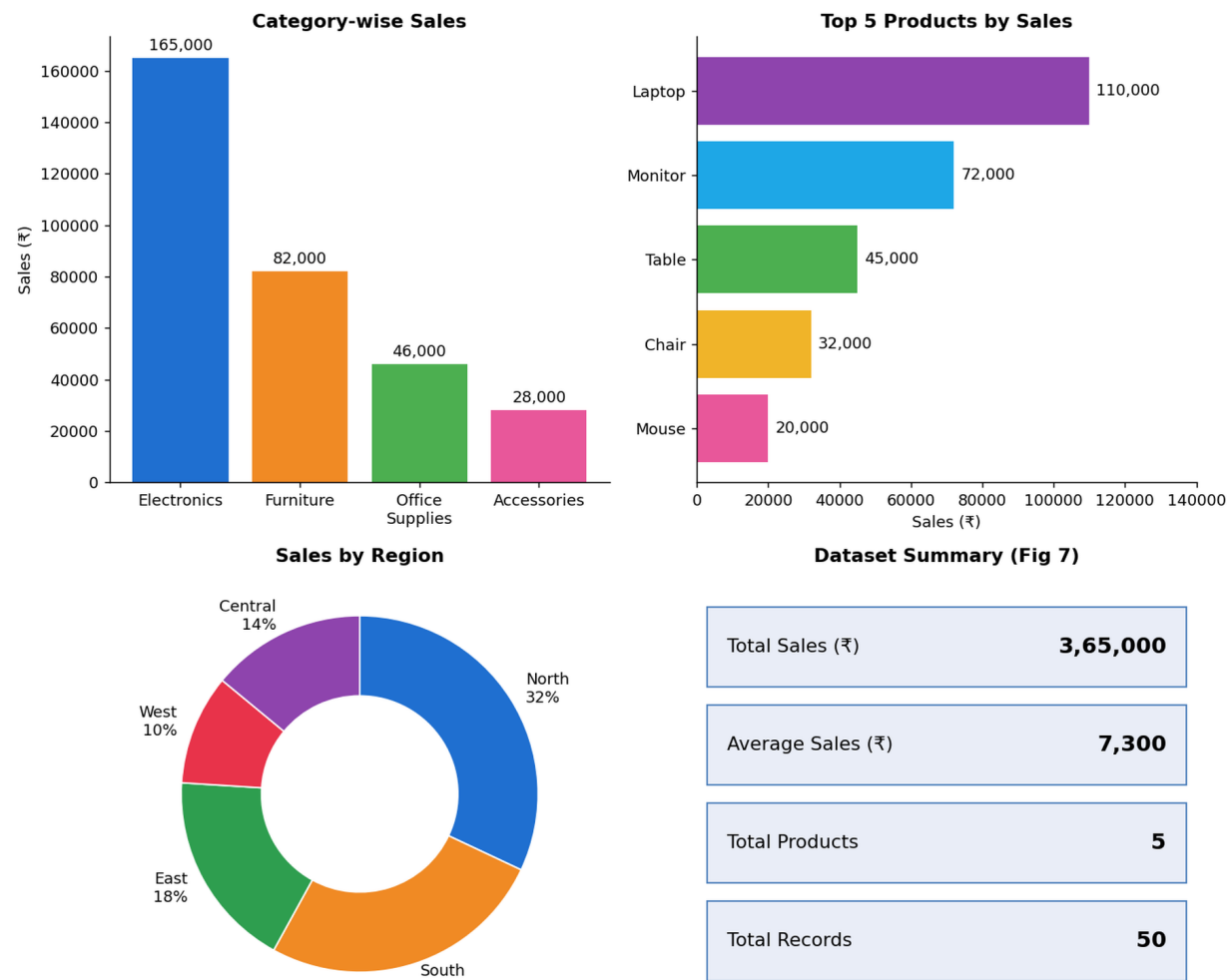
Fig 8: Monthly sales trend (sample output)

4.9 Summary

This chapter explained the complete implementation process of the project, including data understanding, cleaning, analysis, visualization and interpretation. The use of Python and its libraries in Google Colab made the analysis efficient and helped in extracting meaningful insights from the sales data.

CHAPTER 4 AT A GLANCE : VISUAL SUMMARY

The charts below bring together the values shown in the sample visualizations of Chapter 4 (Fig 3) on a single page.



Reading the Charts

CHART	OBSERVATION
Category-wise Sales	Electronics records the highest sales (1,65,000) and Accessories the lowest (28,000).
Top 5 Products	Laptop leads with 1,10,000, followed by Monitor (72,000); Mouse is lowest at 20,000.
Sales by Region	North holds the largest share (32%), followed by South (26%); West holds the smallest (10%).

CHAPTER 5 RESULTS AND DISCUSSION

5.1 Data Quality and Preparation

The data preparation stage focused on checking the available data for missing values, duplicate records, inconsistent formats and other quality issues. The data was organized and prepared so that it could be used consistently for analysis.

5.2 Data Analysis

The prepared data was examined using descriptive and exploratory analysis to understand patterns, comparisons and relationships present in the available information. Observations were interpreted based on the data examined during the internship.

5.3 Category and Attribute Analysis

Where categorical or numerical attributes were available, the data was compared across relevant groups and fields. The analysis helped in understanding differences, distributions and relationships within the available data.

5.4 Visualization-Based Observations

Visualizations provided a clear way to communicate analytical observations. Suitable charts and graphical representations were used to present comparisons, trends, distributions and patterns in an understandable form.

5.5 Learning Outcomes

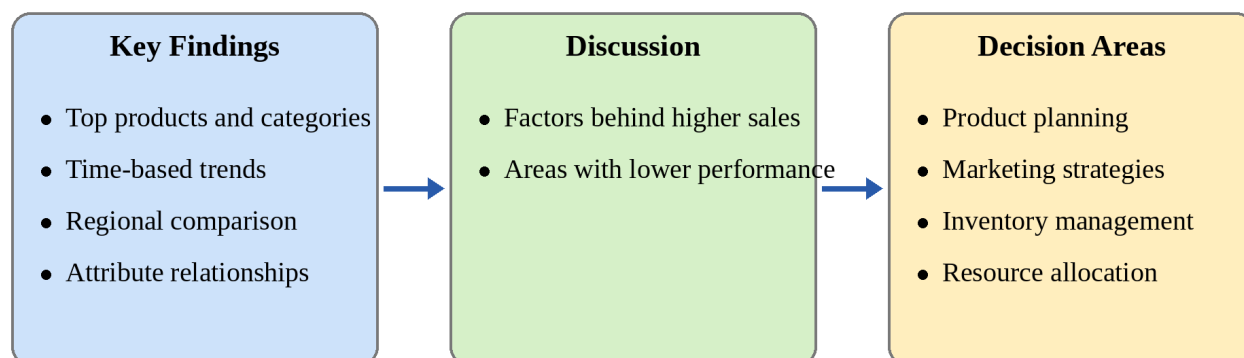
The internship provided practical exposure to the stages of a Data Analytics workflow, including data understanding, preparation, exploratory analysis, visualization and interpretation. It strengthened analytical thinking, technical documentation and the ability to communicate data-driven observations.

CHAPTER 5 AT A GLANCE : ANALYSIS

Analysis Summary Matrix

DIMENSION	MEASURE	VISUALIZATION	WHAT IT REVEALS
Time period	Monthly sales	Line chart	Sales movement across the months of the year
Product	Sales by product	Horizontal bar chart	Top-performing and low-performing products
Category	Category-wise sales	Bar chart	Differences in performance between product categories
Region	Regional distribution	Pie / donut chart	Share of sales contributed by each region
Region and month	Sales across both dimensions	Heatmap	Patterns across regions and months

From Findings to Decisions



The analysis stages, from data quality checks to discussion, are presented in Sections 5.1 to 5.9 of this chapter.

CHAPTER 5 RESULTS AND DISCUSSION (Contd.)

5.6 Key Findings

The analysis revealed several important insights from the sales data, including:

- Identification of top-performing products and categories based on sales volume and revenue.
- Recognition of trends and patterns in sales over different time periods.
- Comparison of performance across regions or customer segments (where applicable).
- Understanding of the distribution and relationship between key attributes.

5.7 Discussion

The findings were interpreted in the context of real-world business scenarios. The analysis highlighted the factors that contributed to higher sales and identified areas with lower performance. These observations can help in making data-driven decisions such as product planning, marketing strategies, inventory management and resource allocation.

5.8 Challenges Faced

During the internship, a few challenges were encountered, such as:

- Handling missing or inconsistent data in the dataset.
- Cleaning and preprocessing data to ensure accuracy.
- Selecting appropriate visualization techniques to clearly represent the insights.
- Interpreting complex patterns in the data.

These challenges helped in improving problem-solving skills and understanding the importance of data quality in analysis.

5.9 Overall Discussion

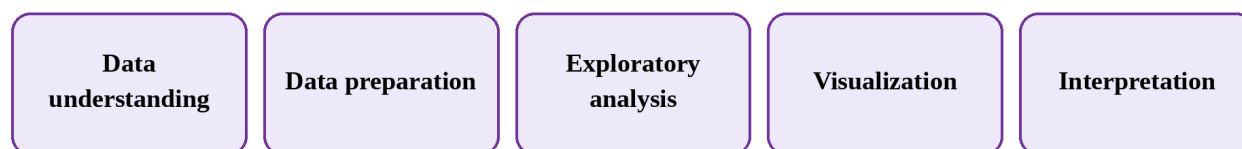
The results and observations from the analysis provided valuable insights into the sales performance and related factors. The use of data visualization made it easier to identify trends, compare attributes and communicate findings effectively. The study demonstrated how analytics can be applied to real-world business problems and support better decision-making.

CHAPTER 5 AT A GLANCE : CHALLENGES

Challenges, Handling and Learning

CHALLENGE	WORKFLOW STAGE	HOW IT WAS HANDLED	LEARNING GAINED
Handling missing or inconsistent data	Data cleaning (3.3, 4.2)	Checked for missing values, duplicates and formatting issues	Importance of data quality in analysis
Cleaning and preprocessing for accuracy	Preprocessing (3.3, 4.2)	Date format conversion, categorical encoding, numeric standardization	Problem-solving and data handling
Selecting appropriate visualization techniques	Visualization (3.5, 4.4)	Line, bar, pie and heatmap charts chosen by purpose	Clear communication of insights
Interpreting complex patterns in the data	Interpretation (4.5, 5.4)	Visualizations reviewed together with analytical summaries	Analytical thinking

Learning Outcomes (Section 5.5)



The internship also strengthened analytical thinking, technical documentation and the ability to communicate data-driven observations.

CHAPTER 6 CONCLUSION AND FUTURE WORK

6.1 Conclusion

The internship project, titled “Sales Data Analysis and Visualization”, provided practical exposure to the complete data analytics workflow using real-world sales data. The project helped in understanding how raw data can be transformed into meaningful information through data cleaning, exploratory analysis and visualization. The analysis revealed important insights such as overall sales trends, top-performing products, category-wise comparison and regional performance. This internship strengthened my analytical skills and improved my ability to interpret data and present the findings in a clear and meaningful manner.

6.2 Overall Internship Outcome

The internship at WIZTECH AUTOMATION SOLUTIONS provided an opportunity to apply academic concepts in a real project environment. It helped me gain hands-on experience in working with data, using Python libraries such as pandas, matplotlib and seaborn, and creating visualizations for analysis. The internship enhanced my skills in data handling, problem solving, analytical thinking and technical documentation. It also improved my confidence in working on end-to-end data science projects.

6.3 Future Work

Based on the learnings and observations from this internship, the following improvements can be considered for future work:

- Extend the analysis with additional sales attributes such as customer segment, discounts and profit margin.
- Develop an interactive dashboard using tools like Tableau or Power BI for real-time exploration of sales trends and comparisons.
- Automate the data-cleaning and reporting workflow using Python scripts.
- Apply advanced statistical techniques or machine learning models, such as demand forecasting, to predict future sales.
- Integrate multiple datasets, such as customer or inventory data, to obtain broader analytical insights.

CHAPTER 6 CONCLUSION AND FUTURE WORK (Contd.)

6.4 Limitations of the Study

Although the analysis provided useful insights, there were certain limitations during the internship:

- The analysis was based only on the available sales dataset, which may not include all business factors such as customer details, profit margins or marketing insights.
- The time period covered in the dataset was limited, which may affect long-term trend analysis.
- External factors such as seasonal demand, market conditions and economic fluctuations were not considered in the analysis.
- The study focused mainly on descriptive and exploratory analysis; advanced predictive modelling techniques were not performed due to dataset limitations.
- Certain attributes that could provide deeper insights, such as customer demographics, discount information and inventory data, were not available in the dataset.

6.5 Summary

This internship project provided a complete learning experience in the field of Data Science. It covered all the major stages of a data analytics workflow, including data understanding, preprocessing, exploratory analysis, visualization and interpretation. The project helped in identifying important sales patterns and presenting them through clear and meaningful visualizations. The knowledge and skills gained during this internship can be applied to future academic and professional projects to solve real-world business problems using data.

Overall, this project strengthened my analytical capabilities, improved my technical skills and increased my confidence in handling data-driven tasks effectively. The insights obtained from the analysis can support better decision-making. In the future, the analysis can be enhanced by incorporating additional data sources, advanced analytics techniques and interactive dashboards.

CHAPTER 6 AT A GLANCE

Future Work and the Limitations It Addresses

FUTURE WORK	LIMITATION ADDRESSED
Extend the analysis with customer segment, discounts and profit margin	Dataset lacked customer details, profit margins, marketing insights, discount information and demographics
Develop an interactive dashboard (Tableau or Power BI)	Enhancement: real-time exploration of trends and comparisons
Automate data cleaning and reporting with Python scripts	Enhancement: a repeatable and faster workflow
Apply advanced statistical or machine learning models such as demand forecasting	Predictive modelling not performed; seasonal demand, market conditions and economic fluctuations not considered
Integrate multiple datasets such as customer or inventory data	Limited time period affects long-term trend analysis; inventory data was not available for deeper insights

Key Takeaways

- Overall sales trends, top-performing products, category-wise comparison and regional performance were the main insights drawn from the analysis.
- The internship gave hands-on experience with pandas, matplotlib and seaborn, and built skills in data handling, problem solving, analytical thinking and technical documentation.
- The knowledge gained can be applied to future academic and professional projects that solve real-world business problems using data.



WIZTECH AUTOMATION SOLUTIONS



Phone : 044-2620 9369 GST : 33DPPPS5321D1ZV UDYAM-TN-02-0249536
New No. 13, 6th Street, A Block, near Anna Nagar East, Roundtana, Anna Nagar, Chennai,
Tamil Nadu 600102

Email : wiztech4automation@gmail.com Web : wiztechautomationsolutions.com

20 AUG 2026

COMPLETION CERTIFICATE

This is to certify that **Mr. A S SUJEETH (REG NO: 110725102252)** studying **B.TECH (AI&DS)** at **"JNN INSTITUTE OF ENGINEERING"** has successfully completed his Internship in **DATA ANALYTICS** in **"WIZTECH AUTOMATION SOLUTIONS"** from **05.08.2026 to 20.08.2026.**

We found the Intern to be hard-working, conscientious and responsible during his tenure with us.

We wish him all the best and success.

With Regards,

DIRECTOR

